place a photon *detector* in front of each slit, the detector's readout would establish with certainty whether the photon went through the left slit or through the right slit, and there'd be no way to erase such definitive information—there'd be no way to recover an interference pattern. But the tagging devices are different because they provide only the potential for which-path information to be determined—and potentialities are just the kinds of things that can be erased. A tagging device modifies a passing photon in such a way, roughly speaking, that it still travels both paths, but the left part of its probability wave is blurred out relative to the right, or the right part of its probability wave is blurred out relative to the left. In turn, the orderly sequence of peaks and troughs that would normally emerge from each slit—as in Figure 4.2b—is also blurred out, so no interference pattern forms on the detector screen. The crucial realization, though, is that both the left and the right waves are still present. The eraser works because it refocuses the waves. Like a pair of glasses, it compensates for the blurring, brings both waves back into sharp focus, and allows them once again to combine into an interference pattern. It's as if after the tagging devices accomplish their task, the interference pattern disappears from view but patiently lies in wait for someone or something to resuscitate it.

That explanation may make the quantum eraser a little less mysterious, but here is the finale—a stunning variation on the quantum-eraser experiment that challenges conventional notions of space and time even further.

## Shaping the Past[17]

This experiment, the *delayed-choice quantum eraser,* was also proposed by Scully and Drühl. It begins with the beam-splitter experiment of Figure 7.1, modified by inserting two so-called down-converters, one on each pathway. Down-converters are devices that take one photon as input and produce two photons as output, each with half the energy ("downconverted") of the original. One of the two photons (called the *signal* photon) is directed along the path that the original would have followed toward the detector screen. The other photon produced by the down-converter (called the *idler* photon) is sent in a different direction altogether, as in Figure 7.5a. On each run of the experiment, we can